A Tree Code for Quantum Error Correction — roadmap handout

Rowan Brad Quni-Gudzinas — QNFO · CWI Summer School poster session, Wed 26 Aug 2026 · full plan: docs/tree-code-roadmap.md in github.com/QNFO/cwi-qec-poster-2026

The claim

Holographic perfect-tensor codes on a truncated Bruhat–Tits tree. Computed thresholds (analytical recursion + exact stabilizer simulation + Monte Carlo, re-verified by a deposited 35-check script): bit-flip **50.0%** (4.6× surface), depolarizing **75.0%** (75×), independent X+Z **17.30%**. **No hardware demonstration yet.**

Where the evidence stands

· Verified computationally, three independent methods; the 35-check verification script is deposited with the guide record.
· Published boundary: under independent errors the qudit family ≈ $2{\times}10^{-4}$ (~55× below surface); the claim targets correlated-failure regimes — test T1

probes exactly this.
· An external no-go was engaged: QEC–Darwinism tradeoff audited on tree spaces; boundary moves to $F_L \approx 0.83$–0.85, residual gap disclosed.
· Self-correction published: a naive p-adic valuation reading of stabilizer codes carries no new content.
· No hardware yet — everything above is classical simulation.

What we test next

#	Test	Where	Disconfirmation criterion
T1	Matched-noise threshold comparison vs surface	Classical, days–weeks	Advantage falls to parity under matched noise
T2	Decoder complexity + full overhead	Classical	Advantage disappears with decoding cost
T3	Random-walk return probability on p-regular trees	Analog/ion, ~8 wk	No change in decay exponent
T4	Clock-rest ultrametricity split (0% vs 29–35%)	Trapped ion, ~8 wk	Split outside predicted range
T5	Independent re-implementation	Community	Reproducibility test

of T1

What we'd like from you

1. An openly specified **correlated-noise benchmark** (model + data) for T1 · 2. decoder-complexity reality check for tree-structured codes · 3. pointers on **perfect tensors for p > 2** · 4. an **independent re-run of T1** — the simulator and protocol are handed over.

DOIs (verified 21 Aug 2026): 10.5281/zenodo.20109836 (architecture) · 10.5281/zenodo.22038733 (guide + verification script) · 10.5281/zenodo.21046993 (qudit QEC) · 10.5281/zenodo.22025544 (trapped-ion testbed) · 10.5281/zenodo.21964674 (QEC–Darwinism audit) · 10.5281/zenodo.21979060 (prime-valuation correction)